

SKANDHAN KARTHIKEYAN

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SKILLS

Programming & Software	- C, C++, Python, MATLAB, Simulink, PyTorch, JAX, ROS2, Linux, Git, SQL
Hardware	- FreeRTOS, PCB design (EasyEDA), I ² C, SPI, UART, CP2102 USB
Tools & Simulators	- CARLA, IsaacSim, MuJoCo, Unity, Inventor, Fusion 360, AutoCAD, P-Scope
Concepts	- TinyML, Signal Processing, Design of Mechatronic Systems, Process Optimization Data Structures & Algorithms, Control and Stability of Aerial vehicles

EXPERIENCE

Course Assistant - TinyML, ESE3600, Penn, USA

Aug 2025 - Nov 2025

- Transitioned development workflows from Arduino IDE to PlatformIO, porting deployments from Arduino Nano 33 BLE to Sseed Studio XIAO ESP32-S3 Sense and reducing on-device inference latency by 15%.

Embedded Systems Intern - Blooming Mill, JSW Steel, India

Jun 2023 - Jul 2023

- Devised a TI C2000 based system for localized sensing, leading to ~10% earlier detection of deformations in steel.

Controls Engineer - Electronics Club, IITM, India

Apr 2022 - Apr 2024

- Designed a custom ESP32-C3 PCB and implemented closed-loop control for VersaGrip, an assistive device to enable fine-motor tasks for paraplegic users. Validated the system through 25+ human-subject trials at R2D2 Lab.

RESEARCH & PROJECTS

Learning-to-Optimize for Model Predictive Control (MPC)

Jan 2026 - May 2026

- Evaluated nonlinear and linearized MPC pipelines in the F1Tenth Gym ROS environment, achieving cross-track errors as low as 0.132 m, at 10m/s top speed on a 3 m-wide racing corridor. Formulated an MPC tuning framework combining Bayesian Optimization and differentiable MPC for controller adaptation under track boundary constraints.

StackX : Autonomous Pick-and-Place Challenge

Aug 2025 - Nov 2025

- Engineered a pick-and-place pipeline integrating perception, planning, and control and executed closed-loop feedback using vision and force sensors to correct object shift, reduced placement misalignment by 4 mm.

RoBA : Real-time 'League of Legends' Challenge

Aug 2025 - Nov 2025

- Implemented closed-loop motion control using encoder feedback and PID for real-time pose control.

LiDAR- and Camera-based Navigation for Lane-Keeping on Semi-structured Roads

Nov 2024 - May 2025

- Developed a lightweight multi-sensor perception pipeline using camera and LiDAR for navigation on semi-structured roads, achieving <0.25 m lateral error. Evaluated performance under varying weather conditions in CARLA.

Delay Compensation in Networked Robotic Systems

Sep 2024 - Feb 2025

- Designed an observer-based mechanism for a mobile robot system during transition, using Delayed Self Reinforcement (DSR), leading to a reduction in cohesion loss by 33%.

Interparticle Force Analysis (Research Internship - North Carolina State University)

Jun 2024 - Aug 2024

- Established a non-invasive technique for interparticle friction evaluation. Improved the computation and temporal efficiency of PeGS by 50% and 35%, respectively. NSF ID: 2104986, titled Mechanics of Granular Materials.

Digital Twin Development, IITM-Accenture Collaboration

Jan 2024 - May 2024

- Developed a Unity-based digital twin for a custom 3D printer achieving ~12% mean absolute temperature prediction error.

EDUCATION

University of Pennsylvania (Penn), Philadelphia, USA

May 2027

Master of Science in Mechanical Engineering & Applied Mechanics (Mechatronics and Robotics Systems)

Indian Institute of Technology Madras (IITM), Chennai, India

Jul 2025

Bachelor of Technology in Mechanical Engineering (GPA : 3.89/4.00)

ACHIEVEMENTS

- Best Presentation, *Young Investigator Workshop*, International Soft Matter Conference (2024)